Data Quality Report

Dirty_customers.csv

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1. Summary

This report reviews the quality of *dirty-customers.csv*, a customer dataset (ID, name, email, signup date, country, spend, segment, active status) covering signups from 1 January 2023 to 19 June 2025. It contains 1,000 rows and 8 columns.

Overall, the data quality is **poor**: every column has at least one issue, and the data isn't ready for analysis without cleaning. The most serious issues are:

- **Missing and placeholder values:** 54 signup dates (5.4%), 62 spend values (6.2%) and 94 emails (9.4%) are missing or filled with placeholders, written in at least 7 different ways (NULL, none, N/A, n/a, -, -999, blank).
- **Inconsistent formats:** signup dates appear in 5 formats, and about 41% of spend values include a $ or USD prefix. 63 emails (6.3%) are malformed.
- **Duplicate customers:** 42 customer IDs appear more than once, adding 43 extra records (4.3%), almost all of them copies of the same customer written with different casing.

2. Dataset Overview

Item	Detail
Source	Dirty-customers.csv
Rows x columns	1000 x 8
Time period	1 Jan 2023 – 19 Jun 2025
Unit of observation	one row per customer
Intended use	The data is meant to analyze customer information, their name, email, signup date, spend, and active status
Distinct Customer IDs	957

signup_date discrepancies:

Group	Count	% of 1,000 rows
ISO	190	19.0%
dd Mon yyyy	188	18.8%
Mon dd, yyyy	187	18.7%
dd.mm.yyyy	193	19.3%
slashes	188	18.8%
Total in a date format	**946**	**94.6%**
`-999`	4	0.4%
Everything else (should be null tokens)	50	5.0%

NULL AND MISSING VALUES:

Null type	Count	% of 1,000
`none`	12	1.2%
`N/A`	11	1.1%
blank	7	0.7%
`-`	7	0.7%
`n/a`	7	0.7%
`NULL`	6	0.6%
`-999`	4	0.4%
Total	**54**	**5.4%**

CUSTOMER_ID COLUMN:
Total: 1000 records

Format	Count	% OF 1000
CUST-0000	720	72%
CUST-00000000	280	28%

FULL NAME COLUMN:
Mixed Cases

EMAIL COLUMN:

Format	Count	% OF 1000
‘string@string.com’	906	90.6%
Invalid format	63	6.3%
Missing/blank email	31	3.1%

COUNTRY COLUMN:

Mixed cases and abbreviations used inconsistently

Spend column:
total rows: 1000
real NaN: 0
Datatype: str

Format	Count	% OF 1000
n/a, N/A	21	2.1%
“”	5	0.5%
NULL	12	1.2%
-999	10	1.0%
-	9	0.9%
none	5	0.5%

DUPLICATE RECORDS:

Metric	Count	% of 1,000
Exact duplicate rows	1	0.1%
Distinct IDs appearing more than once	42	n/a (a count of customers)
Rows belonging to repeated-ID groups	85	8.5%
Extra records beyond one per ID	43	4.3%

3. Summary of Findings

One row per issue. Sort by severity (High first).

Column	Issue	Count / %	Severity	Proposed Action
`signup_date`	Missing or placeholder values (7 spellings)	54 (5.4%)	High	Convert to NULL; exclude from date analysis
`signup_date`	Mixed date formats: ISO 190 (19.0%), dd Mon yyyy 188 (18.8%), Mon dd, yyyy 187 (18.7%),	756 (75.6%) not in ISO format	High	Parse each format explicitly; standardize to YYYY-MM-DD

	dd.mm.yyyy 193 (19.3%), slashes 188 (18.8%)			
spend	Missing or placeholder values (none 5, n/a/N/A 21, NULL 12, -999 10, – 9, blank 5)	62 (6.2%)	High	Convert to NULL; treat -999 as a placeholder, not a number
spend	Mixed formats: $ prefix 205, USD prefix 203, plain 519	408 (40.8%) with a symbol	High	Strip symbols; convert to numeric
customer_id	Repeated IDs: 42 IDs, 43 extra records (1 exact copy)	43 (4.3%)	High	Review pairs; keep one record per ID
customer_id	Two formats: 4-digit (9 characters) vs 8-digit (13 characters)	720 (72%) / 280 (28%)	Medium	Standardize ID format
email	Missing (blank)	31 (3.1%)	Medium	Leave as NULL
email	Malformed	63 (6.3%)	Medium	Repair [at], @@, < > patterns; flag the rest
country	Inconsistent spelling and casing	5 real countries: Brazil, France, Germany, United Kingdom and United States. 14 spellings (13 after lowercasing)	Medium	Map to a standard list
segment	Inconsistent spelling and casing	4 real segment: Enterprise, Consumer, Small business and mid market 15 variants (8 after lowercasing)	Medium	Map to standard categories
is_active	Mixed representations of yes/no	2 real values: True/False 10 variants of 2 values	Medium	Map to True/False
full_name	Stray spaces; all-lowercase or all-uppercase	266 (26.6%); 152 (15.2%)	Low	Trim; standardize casing
spend, signup_date	Stored as text	2 columns	Low	Convert to numeric / date

Severity guide: High = changes your results if ignored. Medium = needs fixing but limited impact. Low = cosmetic.

4. Detailed Findings

4.1 Duplicates

- Exact duplicate rows: [1]
- Duplicates on key column [icustomer_id]: 43, [same data with different case]
- Action: [Review pairs and keep one record per ID after confirming with the data owner.]

[94]:

	customer_id	full_name	email	signup_date	country	spend	segment	is_active
140	CUST-00001104	OMAR WEBER	OMAR.WEBER@CORP.EXAMPLE	2023-11-19	Germany	$268.81	midmarket	Y
564	CUST-00001104	omar weber	omar.weber@corp.example	2023-11-19	Germany	$268.81	midmarket	Y
284	CUST-00001157	ravi Nguyen	ravi.nguyen@corp.example	2023-03-23	Brazil	USD 100.44	ENT	Y
783	CUST-00001157	RAVI NGUYEN	RAVI.NGUYEN@CORP.EXAMPLE	2023-03-23	Brazil	USD 100.44	ent	Y
397	CUST-00001215	Marcus Costa	marcus.costa@example.com	2025-05-16	United Kingdom	540	enterprise	Y
886	CUST-00001215	MARCUS COSTA	MARCUS.COSTA@EXAMPLE.COM	2025-05-16	United Kingdom	540	enterprise	Y
17	CUST-00001273	GRACE FERRARI	GRACE.FERRARI@MAIL.TEST	Apr 19, 2025	u.s.	238.7	small business	FALSE
251	CUST-00001273	Grace Ferrari	grace.ferrari@mail.test	Apr 19, 2025	u.s.	238.7	Small Business	FALSE
546	CUST-00001298	jonas PATEL	jonas.patel@example.com	06.05.2024	u.s.	USD 77.01	enterprise	0
703	CUST-00001298	JONAS PATEL	JONAS.PATEL@EXAMPLE.COM	06.05.2024	u.s.	USD 77.01	enterprise	0
245	CUST-00001302	Sofia haddad	Sofia@@example..com	05/24/2024	Brazil	USD 96.28	SMB	yes
296	CUST-00001302	SOFIA HADDAD	SOFIA@@EXAMPLE..COM	05/24/2024	Brazil	USD 96.28	smb	yes
205	CUST-00001397	RAVI COSTA	RAVI.COSTA@CORP.EXAMPLE	14.07.2024	Brazil	$442.66	consumer	FALSE
931	CUST-00001397	ravi Costa	ravi.costa@corp.example	14.07.2024	Brazil	$442.66	Consumer	FALSE
551	CUST-00001423	MIKAEL BRENNAN	MIKAEL[AT]EXAMPLE.COM	11/19/2024	BR	USD 145.69	small business	1
601	CUST-00001423	Mikael Brennan	Mikael[at]example.com	11/19/2024	BR	USD 145.69	Small Business	1
128	CUST-00001471	PRIYA LINDQVIST	PRIYA.LINDQVIST@MAIL.TEST	Dec 29, 2023	france	16393	smb	false
317	CUST-00001471	PRIYA Lindqvist	priya.lindqvist@mail.test	Dec 29, 2023	france	16393	smb	false
272	CUST-00001476	TOMÁS OKAFOR	TOMS.OKAFOR@EXAMPLE.COM	11 Jul 2023	Brazil	$393.42	mid market	0
745	CUST-00001476	tomás okafor	toms.okafor@example.com	11 Jul 2023	Brazil	$393.42	Mid Market	0
797	CUST-00001588	ana Nguyen	ana.nguyen@example.com	2025-06-08	United States	$318.18	ENT	FALSE
959	CUST-00001588	ANA NGUYEN	ANA.NGUYEN@EXAMPLE.COM	2025-06-08	United States	$318.18	ent	FALSE
196	CUST-00001793	PRIYA OKAFOR	PRIYA.OKAFOR@CORP.EXAMPLE	02.08.2024	france	284	consumer	true
486	CUST-00001793	PRIYA okafor	priya.okafor@corp.example	02.08.2024	france	284	consumer	true
859	CUST-1024	GRACE FERRARI	GRACE.FERRARI@EXAMPLE.COM	2023-04-26	BR	115.96	mid market	TRUE
946	CUST-1024	Grace Ferrari	grace.ferrari@example.com	2023-04-26	BR	115.96	Mid Market	TRUE
447	CUST-1035	AISHA BRENNAN	AISHA.BRENNAN@CORP.EXAMPLE	02/07/2024	United States	236	smb	FALSE
903	CUST-1035	Aisha Brennan	aisha.brennan@corp.example	02/07/2024	United States	236	SMB	FALSE
103	CUST-1037	MIKAEL ODWYER	MIKAEL.ODWYER@CORP.EXAMPLE	04/18/2023	Brazil	USD 189.88	mid-market	0
960	CUST-1037	Mikael ODWYER	mikael.odwyer@corp.example	04/18/2023	Brazil	USD 189.88	Mid-Market	0
150	CUST-1039	OMAR COSTA	OMAR.COSTA@MAIL.TEST	06.03.2024	FR	USD 76.77	ent	false
837	CUST-1039	omar Costa	omar.costa@mail.test	06.03.2024	FR	USD 76.77	ENT	false

4.2 Invalid or impossible values

- **Spend ≤ 0:** Besides the -999 placeholder (10 rows, already counted as missing, so don't count them twice), I found no zero or negative values.
- **Impossible dates:** 0. Everything that matched a format parsed correctly.
- **Future dates:** 0. The latest signup is 19 Jun 2025, so nothing is after report date.
- **Malformed emails:** the 63 malformed and 31 missing.

4.3 Inconsistent formats and categories

Column	Inconsistency	Count

signup_date	5 formats	756 not ISO (75.6%)
spend	$ prefix 205, USD prefix 203	408 (40.8%)
customer_id	4-digit vs 8-digit	720 / 280
full_name	stray spaces / all upper or all lower	266 / 152
country	14 spellings → 5	
segment	15 spellings → 4	
is_active	10 spellings → 2	

Action: standardization rules should be applied

4.4 Wrong data types

- Spend is stored as string datatype
- Signup_date is stored as string datatype
- Action: convert to correct format

Column	Incorrect format	Correct format
signup_date	string	datetime
spend	string	float
is_active	Text	Boolean

4.5 Outliers

- **spend:** method: IQR (upper fence ≈ 1,381). 150 of 938 numeric values (16.0%) exceed it; the median is about 252 and the maximum 171,623. Unknown whether these are errors or genuine large customers.
- Action: need discussion with the client/data owner

4.6 Other issues

- Positive finding: **Date coverage is complete:** all 30 months from Jan 2023 to Jun 2025 appear, with no gaps.

5. Assumptions and Proposed Decisions

- Assumption 1, 'Country'column: DE was read as Germany (ISO code), and GB/UK as the United Kingdom.
- Assumption 2, 'Segment' column: ent is read as enterprise.smb is read as small business
- Dotted dates are read as dd.mm.yyyy and slashed dates as mm/dd/yyyy, because unambiguous rows show this.
- $ and USD are both treated as US dollars.
- Blank cells and the 7 placeholders count as missing.
- 1/0 in is_active mean true/false.

6. Before vs After -- To be completed after cleaning

Metric	Before	After
Total rows	[1000]	[To be completed]
Duplicate rows	[1 exact – 43 extra by ID]	[To be completed]
Total missing values	[147 missing]	[To be completed]

Metric	Before	After
Columns with wrong dtype	[3]	[To be completed]

7. Remaining Limitations

- **Missing values can't be recovered from the data.** 54 signup dates (5.4%), 62 spend values (6.2%) and 31 emails (3.1%) have no value. Any analysis by signup time, spend or email contact will exclude or under-represent these rows.
- **Date order is inferred, not confirmed.** Dotted dates were read as day-first and slashed dates as month-first, because unambiguous rows (day above 12) show this pattern. For the rows where both numbers are 12 or below, the reading depends on that assumption, and only the data's owner can confirm it.
- **Duplicates can't be resolved without a source of truth.** 42 customer IDs appear more than once, and the copies differ in formatting. Without a source system, it's unclear which copy is the correct record.
- **Spend is not safe to total or average yet.** It mixes $ and USD (assumed to be the same currency), contains placeholders, and about 150 of 938 values (16%) lie above the IQR upper fence, with a maximum of 171,623 against a median of about 252. Whether those are errors or genuine large customers is unknown.
- **Malformed emails may not all be repairable.** 63 emails (6.3%) are malformed. Some patterns ([at], @@) can be fixed, but others can't be verified.
- **The data is not current.** The latest signup is 19 Jun 2025, and June 2025 has fewer rows (20) than a typical month (about 30), so it may be incomplete.
- **Columns not to trust yet:** spend (mixed formats, placeholders, outliers) and signup_date (placeholders and inferred date order), until the points above are confirmed.

8. Recommendations

For this dataset (next steps)

- Confirm with the data owner the date order, the currency, and which duplicate copy is correct.
- Clean the data in a separate copy and record every change, so the raw file stays available for comparison.
- Decide whether the spend outliers are errors, and document the decision.

At data entry (prevent the problem)

- Use a date picker or a single required date format (YYYY-MM-DD) instead of free text.
- Use dropdowns for country, segment and is_active, which would remove the 14, 15 and 10 spelling variants.
- Validate emails at entry (reject [at], @@ and anything without a valid domain).
- Require numeric values in spend, with one currency and no symbols.

In the data process

- Use one standard placeholder (a true NULL) instead of the 7 different spellings, and avoid numbers like -999 as placeholders.

- Enforce a unique customer_id with one fixed format, so duplicates are blocked rather than found later.
- Trim spaces and standardize casing automatically on load.
- Add a basic data quality check (null counts, duplicate counts, format checks) that runs on every new file.

9. Appendix

df.info()

```
#   Column          Non-Null Count  Dtype
---  ------      -       -------------  -----
 0   customer_id   1000 non-null   str
 1   full_name     1000 non-null   str
 2   email         969 non-null    str
 3   signup_date   969 non-null    str
 4   country       1000 non-null   str
 5   spend         962 non-null    str
 6   segment       1000 non-null   str
 7   is_active     1000 non-null   str
dtypes: str(8)
```

info() was run on a default pandas read, which converts NULL, blanks and some other strings to NaN, and the later checks use the raw text.

```
total rows:  1000
real NaN:  0


EMAIL INVESTIGATION:
e = df["email"].str.strip()

# 1. Missing: blank
is_missing = (e == "")

# 2. Valid: looks like name@domain.ext
valid_pattern = r"^[^@\s]+@[^@\s]+\.[^@\s]+$"
is_valid = e.str.match(valid_pattern)

# 3. Malformed: not blank and not valid
is_malformed = ~is_missing & ~is_valid

print("missing:  ", is_missing.sum())
print("valid:    ", is_valid.sum())
print("malformed:", is_malformed.sum())
print("total:    ", is_missing.sum() + is_valid.sum() + is_malformed.sum())
```

```
missing:   31
valid:     906
malformed: 63
total:     1000
```

```
sp = df["spend"].str.strip()
placeholders = {"", "null", "none", "-", "-999", "n/a"}
```

```
is_missing = sp.str.lower().isin(placeholders)
num = pd.to_numeric(sp.where(~is_missing).str.replace(r"[^\d.]", "", regex=True), errors="coerce")

q1 = num.quantile(0.25)
q3 = num.quantile(0.75)
iqr = q3 - q1
upper = q3 + 1.5 * iqr

print(q1, q3, iqr, upper)
print((num > upper).sum())
print(num.describe())
```